Figure R4. Qualitative comparison of GPT-5.2 VLM pipelines vs. RAVEL on GLM-Image. Three rare concepts spanning mythology (Goddess Ganga), biology (Red Velvet Worm), and cultural artifacts (Zulu Isicholo), evaluated on GLM-Image, a recent autoregressive model with advanced understanding capabilities. VLM Concept Completion collapses to generic priors: Ganga rendered as a plain woman in a sari, Red Velvet Worm as a generic centipede-like creature, and Isicholo as an Asian conical hat with no cultural grounding. CoT Prompt Expansion and Iterative Verification (K=5) show partial improvements but fail to recover defining attributes: Ganga's multi-armed form and makara vehicle, the Red Velvet Worm's distinctive velvety orange texture and claw-tipped legs, and the Isicholo's characteristic flat-topped silhouette and woven grass construction. Search-enabled CoT improves further but remains inconsistent. In contrast, RAVEL recovers concept-specific attributes in a single pass, with SRD rounds providing targeted refinement toward full structural convergence. Notably, these results are produced on GLM-Image — a model with strong native text understanding — confirming that the performance gap reflects a structural limitation of ungrounded LLM pipelines rather than the generative model's prompt comprehension capacity.

**Reference images are provided for comparison purposes only and are not used during generation.*